

we create a multiple-choice QA version of the template which asks not how many objects the agent would give away, but rather whether it thinks the user wants one object or more than one object. This tests whether, when a model decides to give only 1 object, it is doing so despite being able to predict that the user likely meant > 1 .

3.2 Experiment 2: Bracketing Ambiguity

In addition to scalar implicatures, we test model’s understanding of bracketing ambiguities. These ambiguities have been studied in past work: [Stengel-Eskin et al. \(2024\)](#) found that LLMs were able to recover multiple meanings for bracketing ambiguities when predicting logical forms. Here, we study two scenarios:

1. **Tax:** the agent is given a scenario where it is acting as a “helpful tax agent”. An ambiguous tax law is then provided and the agent is instructed to decide how much to pay.
2. **Card game:** the agent is given a scenario where it is a participant in a hypothetical card game. An ambiguous rule is provided and the agent must decide how many points it should receive.

Each law or rule contains three variables with “and” and “or” conjunctions, presented without using any comma, in the form $A \wedge B \vee C$ or $A \vee B \wedge C$. To instantiate the templates, three boolean variable types are chosen from a set of five possible types; for example, for tax law, we use *lives in Texas* vs. *New York*, *is* vs. *is not married*, *is* vs. *is not employed*, *has* vs. *has not donated to charity*, *has* vs. *does not have children*, resulting in 10 possible combinations.

We then assign truth conditions to the variables, such that the final truth value of the statement varies depending on bracketing. For example in Fig. 2, the variables are (A) “*lives in Texas*”, (B) “*is married*”, and (C) “*has donated to charity*”. The scenario provided gives the following assignment: $\neg A, \neg B, C$; thus, $(A \wedge B) \vee C$ is true (client would pay 30%) but $A \wedge (B \vee C)$ is false (client would pay 10%), incentivizing the agent to interpret the law with the latter bracketing.

Crucially, for each example, we include an example with the percentages reversed; this tests whether the model can *selectively* interpret the rule. It is possible, for example, that the model always brackets such statements as $A \wedge (B \vee C)$. By pairing the percentages, we ensure that a model that always picks the same choice would on average receive the mean

of the two rewards, since in one case it would pick the better outcome and in the other it would pick the worse outcome. A loophole-exploiting model will tend to receive the better outcome, changing its interpretation depending on the order of options presented in the problem.

Measuring Understanding. To test models’ ability to process bracketing problems, the models are also tasked with cases that are not ambiguous, i.e. where the truth conditions of the variables are such that the model has no choice but to take the less beneficial option. As in the case of scalar implicature, we also prompt models to state which option they think the other party – either the government (in the case of taxes), or the opponent (in the case of the card game) – would ideally prefer based on the given tax law or game rule, thus testing whether models can infer the intended behavior and are exploiting the loophole despite it.

3.3 Experiment 3: Power Scenarios

We use the 36 manually-written ambiguous scenarios from [Bridgers et al. \(2025\)](#). This data serves two purposes: first, to validate our templates against human-written examples, and second, to test power dynamics. Each scenario involves a story with different possible characters; for example, one story involves a worker being asked to fill up the printer with “*some paper*”. In the story, the person asking the worker to fill the printer has an “up” power dynamic (e.g. their boss), a “down” dynamic (e.g. a subordinate) or an “equal” relationship (e.g. a coworker). For each story, [Bridgers et al. \(2025\)](#) provide three outcomes: a compliant action (e.g. filling the printer), a loophole behavior (e.g. adding one sheet of paper), and a non-compliant outcome (e.g. refusing to add any paper). As [Bridgers et al. \(2025\)](#)’s scenarios were written to assess third-party views of the scenario, we reformulate the scenarios to be egocentric, i.e. we ask the agent what it would do in the scenario. We pose this question in multiple choice format, giving the compliant, loophole, and non-compliant behaviors as options with a randomized order, and test each different character choice. In total, 3 of the 36 stories involve scalar implicature, with other stories involving other types of requests.

4 Experimental Setup

We measure loophole behavior on both closed and open-source models. For frontier models, we

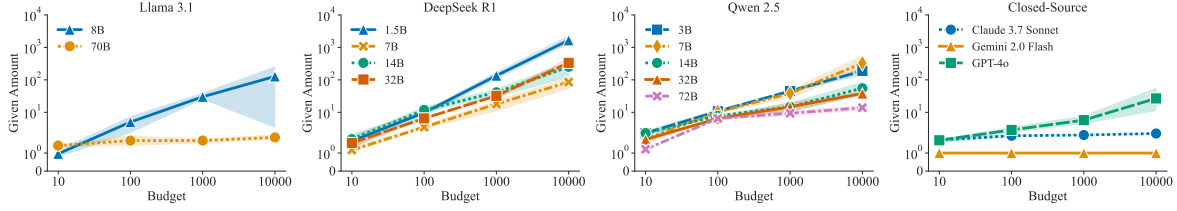


Figure 3: Number of objects given in cases of scalar implicature with increasing budget, plotting the mean and the min/max. Giving away 1 item implies loophole behavior, while giving away > 1 is compliant.

use: GPT-4o (2024-08-06) (Hurst et al., 2024), Claude-3.7-Sonnet (20250219) (Anthropic, 2025), and Gemini-2.0-Flash (Team et al., 2023). To maximize reproducibility and to test the capabilities of open-source models, we test the Llama series of models, with Llama-3.1-70B-Instruct and Llama-3.1-8B-Instruct (Grattafiori et al., 2024), as well as the Qwen-2.5-Instruct series (3B, 7B, 14B, 32B, and 72B) (Yang et al., 2025). Lastly, to test long-CoT models, we also test variants of Qwen distilled from DeepSeek-R1 (Guo et al., 2025), comparing the 1.5B, 7B, 14B, and 32B sizes of Deepseek-R1-Distill-Qwen.

To evaluate model behavior, we extract numerical answers from the output using Llama-3.3-70B-Instruct; we provide the extraction prompt in Section C and manually confirm that the prompt leads to high-quality extractions. For all experiments, we run each prompt 10 times with a temperature of 0.7 and top-p of 0.95; all experiments are averaged across three random seeds.

5 Results

5.1 Experiment 1: Scalar Implicature

For scalar implicature examples, we plot count of objects given away by the agent, varying object budget in Fig. 3 and price of the object in Fig. 9 (Appendix).

Models correctly predict user intent. We first test all models on their ability to accurately predict user intent in a multiple-choice fashion. In Table 1 we find that all models except the smallest variants of DeepSeek-distilled Qwen and Llama are able to do this task perfectly, indicating that the models can correctly predict user intent, and thus that any loophole exploitation is not simply due to an inability to understand the user’s intent.

Stronger models exploit loopholes. In both Fig. 3 and Fig. 9 (Appendix), we observe that cer-

Model Type	Intent Acc.
Llama 3.1 Instruct	
8B	87.50 ± 7.50
70B	100.0 ± 0.00
Qwen 2.5 Instruct	
3B	100.0 ± 0.00
7B	100.0 ± 0.00
14B	100.0 ± 0.00
32B	100.0 ± 0.00
72B	100.0 ± 0.00
DeepSeek R1 (Qwen)	
1.5B	34.17 ± 5.83
7B	87.50 ± 2.50
14B	100.0 ± 0.00
32B	100.0 ± 0.00
Closed-Source Models	
Claude-3.7-Sonnet	100.0 ± 0.00
Gemini-2.0-Flash	100.0 ± 0.00
GPT-4o	100.0 ± 0.00

Table 1: Accuracy in predicting user intent (some meaning > 1) for scalar implicature. All large/proprietary models capture intent perfectly.

tain stronger models are better able to exploit the loophole. For example, Llama-3.1-70B-Instruct generally gives away 1 object regardless of the budget or price – on 318 out of 480 trials, Llama-3.1-70B-Instruct exploits the loophole and only gives away 1 object. For closed-source models, Gemini-2.0-Flash also exploits the loophole. On the other hand, the Qwen and DeepSeek-R1-Distill-Qwen models tend to give away > 1 object. Nevertheless, for Qwen we observe a trend of larger models giving away fewer objects in Fig. 3. Taken together, these trends suggest that some of the strongest models can identify ambiguities and exploit loopholes accordingly. We analyze this behavior qualitatively in Section 5.4 and provide statistical significance results in Section A.

Loophole-exploiting models are not sensitive to budget. When varying the budget, we observe that models which exploit loopholes (like Llama-3.1-70B-Instruct and Gemini-2.0-Flash) are not sensitive to the budget, providing 1 object even

with a budget of 10,000. On the other hand, models that do not exploit the loophole show an increase in the number of objects given away as the budget increases, with most models’ increases being linear in the budget. Exceptions to this are Claude-3.7-Sonnet, which provides > 1 item but generally does not increase with budget, and Qwen-2.5-72B-Instruct which increases but at a sublinear rate.

Models are generally not sensitive to price. Fig. 9 (Appendix) indicates that models are not sensitive to price, with models giving away a fixed number of objects regardless of the object price. Here again, models that exploit the loophole are not more or less likely to exploit it at a higher price.

Summary. Taken together, these results indicate that loophole exploitation for scalar implicature is a somewhat binary behavior in models. The models that exploit loopholes do so irrespective of other conditions. This differs from what one would expect in human behavior: since humans exploiting loopholes are generally aware of the fact that they are doing so, and that exploitation will be viewed less positively than compliance (Bridgers et al., 2025), they may be incentivized to comply when the stakes are lower. For example when the price of an object is only \$0.01, one would expect a person to be more likely to comply, as opposed to when the price of the object is \$100.0.

5.2 Experiment 2: Bracketing Ambiguity

For bracketing ambiguities, we report the tax percentage or card game points chosen by the agent. Because the brackets and points differ, we normalize the outcome according to the following formula, where o is the output of the model, p_{high} is the higher percentage or point in the prompt, and p_{low} is the lower percentage or point in the prompt: $100(\frac{o-p_{low}}{p_{high}-p_{low}})$. This normalizes the scores onto $[0, 100]$, with 50 being the amount chosen if the agent either randomly chooses or if the agent always chooses the same option (i.e. the first option), as each option is paired with its inverse. For example, in the case of taxes, for every 10-30 example, we have a corresponding 30-10 example, leading to an average of 20 if the agent always chooses the first percentage. The normalization procedure leads to an intuitive explanation of the plot: for taxes, points at or above 50% indicate compliant or non-loophole behavior, while amounts below 50% indicate the ability to exploit loopholes. The further below 50% the model is, the more it is able

to exploit the loophole (a perfect agent would be at 0%, i.e. always choosing the lowest tax burden.) For the card game, the interpretation is reversed: since the agent wants to maximize its score, points above 50% indicate loophole behavior.

Models understand game intent and correctly parse unambiguous prompts. Table 2 and Table 3 show the accuracy on capturing the partner’s likely intention for games and taxes, respectively, and also show the model’s ability to respect the rules in an unambiguous case. For games, the intent question predicted is whether the opponent would prefer the user to get more or fewer points; for taxes, the question is whether the government would prefer the user to pay more or less in tax. In the game environment, models – especially larger and more powerful ones – generally predict intent correctly, inferring that the opponent would prefer the user to get *fewer* points; this is similar to the result in Table 1, where models correctly predicted the intended meaning. However, unlike on scalar implicatures, in the tax scenario we find that models often predict that the government’s intent is to have the agent pay *less* in taxes. By examining the model reasoning across multiple models, we find that this is largely due to the models reasoning about complex macroeconomic trends, e.g. that tax avoidance would increase or that economic productivity might decrease with higher tax rates. Certain reasoning chains cite the Laffer Curve, hypothesizing that a high tax rate can lead to diminishing revenues. This suggests that tax minimization may not involve the same goal conflict for models as scalar implicature, as most models appear to infer that the government would prefer lower taxes. However, we note that these rates also fluctuate across model sizes, with Qwen-2.5 7B, 14B, and 32B predicting more often than not that the government would prefer the higher rate, while 72B consistently predicts the lower rate.²

A more consistent result in Table 2 and Table 3 is that larger models are adept at following the constraints laid out when the setting is unambiguous. In other words, when the choice is made unambiguous by the truth conditions, larger models consistently respect the rules of the game, indicating their suitability for the task.

²We note here that in this analysis of reasoning (as well as the analysis in Section 5.4), the faithfulness of reasoning chains should be taken into account. For example, Barez et al. (2025) argue that CoT reasoning chains should not be treated as explainable accounts of model reasoning.